B2.2 The challenges of size

Summary

When organisms become multicellular, the need arises for highly adapted structures including gaseous exchange surfaces and transport systems, enabling living processes to be performed effectively.

Underlying knowledge and understanding

Learners should be familiar with the role of diffusion in the movement of materials in and between cells. They should also be familiar with the human gaseous exchange system.

Common misconceptions

Learners have a view that the slow flow of blood in capillaries is due to the narrow diameter, when in fact it is a function of the total cross-sectional area of the capillaries (1000 times greater than the aorta). When explaining the importance of the slow flow of blood in allowing time for exchange by diffusion, this misunderstanding should be considered.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM2.2i	calculate surface area : volume ratios	M1c
BM2.2ii	use simple compound measures such as rate	M1a and M1c
BM2.2iii	carry out rate calculations	M1a and M1c
BM2.2iv	plot, draw and interpret appropriate graphs	M4a, M4b, M4c and M4d

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	**To include**	**Maths**	**Working scientifically**	
B2.2a explain the need for exchange surfaces and a transport system in multicellular organisms in terms of surface area : volume ratio	calculation of surface area, volume and surface area : volume ratio, and reference to diffusion distances	M1c	WS1.4d, WS1.4e, WS1.4f, WS2a, WS2b, WS2c, WS2d	Investigating surface area : volume ratio using hydrochloric acid and gelatine cubes stained with phenolphthalein or other suitable pH indicator. (PAG B8)
B2.2b describe some of the substances transported into and out of a range of organisms in terms of the requirements of those organisms	oxygen, carbon dioxide, water, dissolved food molecules, mineral ions and urea			
B2.2c describe the human circulatory system	the relationship with the gaseous exchange system, the need for a double circulatory system in mammals and the arrangement of vessels			Modelling of the human circulatory system.
B2.2d explain how the structure of the heart and the blood vessels are adapted to their functions	the structure of the mammalian heart with reference to the cardiac muscle, the names of the valves, chambers, and blood vessels into and out of the heart, the structure of the blood vessels with reference to thickness of walls, diameter of lumen, presence of valves		WS2a, WS2b, WS2c, WS2d	Investigating heart structure by dissection. Investigation of a blood smear using a light microscope. (PAG B1) Modelling of blood using sweets to represent the components.
B2.2e explain how red blood cells and plasma are adapted to their transport functions in the blood			WS2a, WS2b, WS2c, WS2d	Examine the gross structure of blood vessels using a light microscope. (PAG B1) Investigating of the elasticity of different blood vessels using hanging masses.

B2.2f	explain how water and mineral ions are taken up by plants, relating the structure of the root hair cells to their function			WS2a, WS2b, WS2c, WS2d	Examination of root hair cells using a light microscope. (PAG B1) Demonstration of the effectiveness of transpiration by trying to suck water from a bottle using a 10m straw. (PAG B8) Investigation of the position of the xylem/phloem in root, stem and leaf tissues using a light microscope. (PAG B1) Interpretation of experimental evidence of the movement of dissolved food materials in a plant. (PAG B1, PAG B8) Examining the position of the phloem in root, stem and leaf tissues using a light microscope. (PAG B1)
B2.2g	describe the processes of transpiration and translocation	the structure and function of the stomata		WS2a, WS2b, WS2c, WS2d	Measurement of plant stomatal density by taking an impression of the leaf using clear nail varnish or spray-on plaster. (PAG B1, PAG B6, PAG B8)
B2.2h	explain how the structure of the xylem and phloem are adapted to their functions in the plant				
B2.2i	explain the effect of a variety of environmental factors on the rate of water uptake by a plant	light intensity, air movement, and temperature	M1a, M1c, M1d	WS2a, WS2b, WS2c, WS2d	Interpreting experimental evidence of investigations into environmental factors that affect water uptake. (PAG B6, PAG B8)
B2.2j	describe how a simple potometer can be used to investigate factors that affect the rate of water uptake	calculation of rate and percentage gain/loss of mass	M1a, M1c, M1d, M2g, M3d, M4a, M4b, M4c, M4d	WS1.2b, WS1.2c, WS1.2e WS1.3a, WS1.3b, WS1.3c, WS1.3d, WS1.3e, WS1.3f, WS1.3g, WS2a, WS2b, WS2c, WS2d	Investigation of transpiration rates from a plant cutting. (PAG B6, PAG B8) Work out the rate of transpiration in volume of water/time. (PAG B6, PAG B8)